The Ångström Problem

Why structural prediction can tell you *where* a mutation sits but not *whether* a drug will still work — and what to build instead.

> **The short version.** Tier 2 asks a docking score to resolve differences of `0.02–0.56` kcal/mol. The instrument's own error bar is `1.0–2.19` kcal/mol. We are reading below the noise floor. This is not a bug, a tuning problem, or an engineering failure — it is a mismatch between the question and the instrument, and it has known alternatives.

PART 1

What we are actually trying to do

A patient has a cancer with a specific mutation. Standard-of-care has failed, or the mutation is not in any database. The oncologist needs a defensible next step.

Tier 1 answers this by retrieval: it searches curated evidence (CIViC), clinical trials, and literature for what is *documented* about this variant. Every answer carries a citation. This is the safe, primary path.

Tier 2 exists for when Tier 1 finds nothing. There is no citation to give, so instead we try to *compute* something from first principles — from the physical structure of the protein and the drug.

That is the whole premise, and it is a reasonable one. The question is what, precisely, we can compute.

PART 2

The chain of assumptions behind the scenes

Tier 2's original design rests on a chain of four inferences. Each looks obvious. Each is an assumption, and three of them are conditional.

THE INFERENCE CHAIN	Every arrow is a claim that must hold
mutation ↓ structure change	A single amino-acid substitution changes the protein's shape. MOSTLY FALSE A point mutation leaves the fold essentially unchanged. This is well established — it is precisely why our own spec forbids re-folding the mutant. The change is local: one side chain.
structure change ↓ binding change	The altered shape changes how well the drug binds. ONLY IF NEARBY True when the residue contacts the drug. A residue 16 Å away cannot alter the contact surface, because it is not part of it.
binding change ↓ clinical resistance	Weaker binding means the drug stops working. ONE OF MANY ROUTES Binding loss is one mechanism of resistance among at least five. See Part 3.
binding change ↓ we can measure it	Docking can quantify the difference. NOT AT THIS SCALE The difference we need is smaller than the method's error. See Part 5.

THE TEACHING POINT

None of these arrows is *wrong*. Each is *conditional*. The design treated them as unconditional, so the pipeline was scoped to answer a general question using a method that only answers a specific one.

PART 3

The biology: what actually causes resistance

When a targeted therapy stops working, the cause is one of several distinct mechanisms. Only the first is a binding-affinity problem.

RESISTANCE MECHANISMS

Structural docking addresses row 1 only

MECHANISM	EXAMPLE	WHAT CHANGES	DOCKING?
Steric / chemical interference	`EGFR T790M`	The residue physically contacts the drug; substitution alters the contact surface	YES
Conformational shift	`BRAF V600E`	Equilibrium between active and inactive protein states moves	NEEDS ENSEMB
Allosteric	`KIT D816V`	A distant residue stabilises a state the drug cannot bind	NEEDS DYNAMI
Bypass pathway	`MET amplification`	A different protein takes over the signalling role	DIFFERENT PR
Expression / efflux / plasticity	—	Target abundance, drug transport, or cell identity changes	NOT STRUCTUR

CONSTRAINT 1 · SELECTION BIAS IN THE TEST SET

Our nine gold-standard cases were chosen because they are *clinically famous*. Clinical fame does not correlate with structural tractability. The set spans all five rows above, so most of it was never answerable by this method — the validation was measuring a mismatch, not an implementation.

PART 4

The physics: what "binding affinity" is and how it is computed

Binding affinity is a free energy — ΔG, in kcal/mol. More negative means tighter binding. What we want is $\Delta\Delta G$: the *difference* between how the drug binds the normal protein and how it binds the mutant.

How docking estimates it

AutoDock Vina samples thousands of orientations of the drug inside the binding pocket and scores each with an empirical function — a formula fitted to known complexes, weighting hydrophobic contact, hydrogen bonds, and steric clash. It returns the best score found.

To get $\Delta\Delta G$ we run this twice, once per protein form, and subtract. **Subtracting two noisy estimates gives an answer noisier than either.**

The three instruments in play

METHODS AND THEIR PUBLISHED ACCURACY — Figures from the project's own research review

METHOD	QUESTION IT ANSWERS	REPORTED ACCURACY	AVAI
AlphaMissense	Is this variant likely pathogenic?	90% precision on ClinVar	ALW
mCSM-lig	How does this mutation change drug affinity?	ρ 0.67 · 0.73–0.77 within 5 Å re-test: PCC 0.63, RMSE 2.19	NEE
AutoDock Vina	How well does this drug fit this pocket?	evaluated and *failed* on resistance mutations	ALW

TWO THINGS WORTH NOTICING

AlphaMissense answers a different question. It predicts whether a variant damages the protein — not whether a particular drug still binds. A variant can be highly pathogenic and leave drug binding untouched. Conflating the two would be a category error.

mCSM-lig is the right tool and it agrees with us. Its accuracy is ρ 0.627 overall but ρ 0.674 for mutations *within 5 Å of the ligand* — the headline figure is the near-ligand case. The method's own literature says the signal lives near the pocket, independently confirming the distance threshold we reached from our measurements. On independent test sets it drops materially (R ≈ 0.43 on a TKI set).

Vina has been evaluated on this exact task, and failed it. This is stronger than "no figure exists": a direct 2026 assessment found scoring functions including Vina failed to identify drug-resistant mutations, on which dedicated mutation-based methods (mCSM-lig, PremPLI, Rosetta flex-ddG) succeeded. Our primary-traffic method has been measured against our question and did not answer it.

PART 5

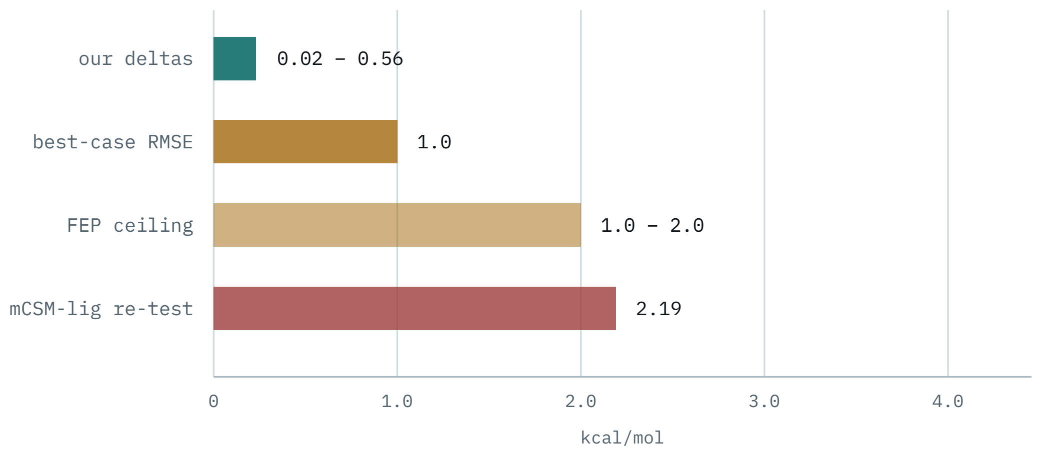
The decisive constraint: signal below the noise floor

This is the single most important slide. Everything else is detail.

Below, on one scale in kcal/mol: the differences we measured across the validation set, against the error bars of the instruments measuring them.

SIGNAL VS. INSTRUMENT ERROR

kcal/mol · same scale throughout



The teal bar is everything we are trying to detect. The bars beneath it are how wrong each method is expected to be. No threshold recovers a signal smaller than the error of the instrument reading it.

CONSTRAINT 2 · RESOLUTION

We are using a ruler marked in centimetres to measure differences in microns. This is why the gefitinib delta moved from `+0.556` to `+0.313` when we changed nothing but receptor preparation — with a fixed random seed. We were reading noise, and noise moves.

The construction flaw that makes it circular

There is a second, sharper problem. We build the mutant protein by swapping one side chain on the wild-type backbone — deliberately, because re-folding a point mutant just adds prediction error to an essentially identical structure.

But that means our two structures differ by *exactly one side-chain rotamer*, by construction. Any effect that works through backbone motion or conformational change is **definitionally invisible** — we removed it from the model before measuring.

CONSTRAINT 3 · THE BLIND SPOT MATCHES THE FAILURES EXACTLY

BRAF V600E and KIT D816V did not fail because docking is imprecise. They failed because their mechanisms are conformational and allosteric, and we built those out of the representation. The method's blind spot is precisely the set of cases it got wrong.

The trap that produces a confidently wrong answer

Some drugs work by forming a *covalent bond* to the protein. Osimertinib bonds EGFR Cys797. Both our scoring methods are non-covalent — their scoring functions contain no term for bond formation, so they evaluate only the reversible pose.

For **EGFR C797S**, resistance *is* the loss of that cysteine. The reversible pose still fits the pocket, so a non-covalent score can come back unchanged and report **retained binding for a textbook resistance case** — wrong, and in the most dangerous direction.

CONSTRAINT 4 · THE INVALID CONTROL

This also invalidates one of our two pre-committed positive controls: EGFR T790M/osimertinib was scoring a covalent drug with non-covalent methods. It was never a valid test as constructed. The pipeline now gates covalent drugs before scoring and returns `mechanism_invalidated` instead of a number.

The silent error nobody sees

Kinases exist in distinct conformational states — `DFG-in` (active) and `DFG-out` (inactive). Type-II inhibitors such as **imatinib, nilotinib, and sorafenib bind only the DFG-out state.** AlphaFold predominantly returns DFG-in.

So docking imatinib into an AlphaFold model can mean docking it into a pocket it does not bind — producing a confident number that is meaningless, with no error raised. Our research review names this "the single largest silent error source", and notes it applies to *exactly* the ABL1 / KIT / ALK / BRAF targets in our validation set.

GOOD NEWS

This one is solvable. Tools exist (KLIFS, Kincore) that classify kinase conformational state from a structure with near-perfect accuracy — one benchmark mislabelled `17 of 8,826`. We simply do not check it yet.

PART 6

The full constraint inventory

Everything that limits Tier 2, grouped by whether it is fixable and at what cost.

CONSTRAINTS

Physics and biology are fixed; the rest are engineering

CLASS	CONSTRAINT	FIXABLE?
Biology	Most resistance is not binding-mediated	FIXED
Biology	Point mutations barely change the fold; effects are local	FIXED
Physics	Docking error exceeds the signal by 2–40×	CHANGE METHOD
Physics	Rigid receptor; conformational ensembles not represented	ENSEMBLE DOCKING
Data	Not every protein has an experimental structure	ALPHAFOLD (apo)
Data	Structures may not resolve the mutated residue	SCAN MORE ENTRIES
Data	The bound ligand is usually not the drug being evaluated	DISCLOSE IT
Data	AlphaFold models are apo — no pocket to measure against	DOCK A REFERENCE
Data	Kinase conformational state unchecked (DFG-in/out)	KLIFS / KINCORE
Method	mCSM-lig requires the drug to be a known PDB ligand	INHERENT
Method	Covalent inhibitors scored by non-covalent methods	GATE IT
Data	UniProt and PDB residue numbering can disagree	SIFTS MAPPING
Engineering	Public API rate limits cost whole cases	RETRY + CACHE
Engineering	Tooling gaps (meeko rejects mutated residues)	UPSTREAM FIX
Epistemic	A cutoff fitted on the test set cannot be validated by it	DISCIPLINE ONLY
Epistemic	"Not measured" must never render as "measured negative"	ALREADY ENFORCED

PART 7

What we measured

The pipeline runs end to end. These are real numbers from real structures, not estimates.

MUTATION-TO-POCKET DISTANCE — Measured from experimental coordinates

CASE	DISTANCE	BAND	KNOWN MECHANISM
EGFR T790M	3.5 Å	IN CONTACT	gatekeeper — direct contact
ABL1 T315I	3.5 Å	IN CONTACT	gatekeeper — direct contact
ALK G1202R	3.7 Å	IN CONTACT	solvent front — direct contact
ALK I1171T	6.9 Å	POCKET-ADJACENT	pocket-lining
EGFR C797S	10.1 Å	DISTANT	covalent-site loss
BRAF V600E	12.2 Å	DISTANT	conformational activation
KIT V560G	20.0 Å	DISTANT	juxtamembrane

Every contact-mediated mechanism lands `in_contact`. Every conformational or allosteric one lands `distant`. The measurement separates the two cleanly.

> CAVEAT WE MUST STATE, EVERY TIME
>
> The distance is measured to the ligand that was *co-crystallised in that structure*, which is usually **not** the drug being evaluated. EGFR structure 8A27 contains ligand KY9 — so gefitinib and osimertinib both read 3.5 Å, identical, because neither number is about those drugs specifically.
>
> The claim is therefore *pocket-level*, not drug-specific: it locates the mutation relative to the occupied binding site. For inhibitors competing for the same ATP pocket that is the relevant geometry — but it must be said as what it is.

AND THE BINDING DELTA ITSELF

Of nine variant-drug pairs, one produced a delta inside the method's applicable range, and it came out `uncertain`. Pass rate against the pre-committed criteria: **0 of 9.** Both hard-required positive controls failed. We did not move the threshold.

PART 8

The options

Every constraint above has a response. They differ enormously in cost and in how much they actually buy.

A · Retrieval, not prediction

HIGHEST VALUE

Do not compute ΔΔG — look it up. Platinum, MdrDB, and the TKI resistance literature contain *measured* affinity shifts for these exact variant-drug pairs. This turns an intractable physics problem into a retrieval problem, which is what Tier 1 already does well.

COST	Data ingestion; no new science
BUYS	Real measured values with citations
LIMIT	Only covers variants somebody has measured

B · Mechanism classification

BUILT

Ask what structure can answer: which mechanism class is geometrically available here — contact, pocket-reshaping, or neither? Tells a clinician what *kind* of problem they face, which is actionable even without a number.

COST	Done — running on 8 of 9 cases
BUYS	A defensible claim available today
LIMIT	Not a response prediction, and must never be sold as one

C · Calibration

NEXT INCREMENT

Train on MdrDB/Platinum: features (predicted ΔΔG, AlphaMissense, distance, mutation class) → empirical resistance probability, wrapped in conformal prediction. This does not make docking accurate; it *learns how inaccurate it is* and reports honest intervals.

COST	Dataset access + modelling work
BUYS	Calibrated uncertainty instead of a bare number
LIMIT	Still bounded by the underlying signal

D · Conformational state check

CHEAP WIN

Classify DFG-in/out before docking and refuse — or correct — when the structure is in the wrong state for the inhibitor class. Removes the largest silent error, and the classifiers are near-perfect on crystal structures.

COST	Integrate KLIFS or Kincore
BUYS	Eliminates confident-but-meaningless results
LIMIT	Kinases only

E · Free energy perturbation

FUNDED PHASE

The physically correct instrument, and better than we first credited. A consensus of alchemical free energy and Rosetta reached **RMSE 0.62 kcal/mol, r = 0.71** on 144 experimental Abl ΔΔG values; a separate study computed resistance or sensitivity at **88% accuracy** across 144 Abl point mutations — demonstrated on exactly this kinase-resistance problem. Open implementations exist (OpenFE, pmx).

COST	~1 GPU-day per mutation-drug pair
BUYS	An answer at the right resolution, on this exact problem class
LIMIT	Viable only for a small curated set — e.g. pre-computed demo cases

F · Ensemble docking

PARTIAL

Dock against multiple conformations rather than one static structure, recovering some of the conformational effects the single-pose approach removes by construction.

COST	N× compute; needs a conformer source
BUYS	Partial coverage of mechanism row 2
LIMIT	Does not fix the underlying resolution problem

RECOMMENDATION

A and B are the product. Retrieval where measurements exist, mechanism classification where they do not, and an explicit statement of which regime you are in. **D is the cheapest real improvement.** C is the credible next increment. E is a funded-phase item.

PART 9

What we can and cannot say

The line below is the whole governance question. Everything on the left is backed by a computation we can show. Everything on the right is not.

Supported today

- This mutation is valid and sits at this position in this canonical isoform.
- Its AlphaMissense pathogenicity score is *X*, using the published thresholds.
- We sourced this experimental structure, with this confidence at this residue.
- These drugs target this protein, from these databases.
- The mutation sits *N* Å from the pocket occupied by ligand *L* in that structure.
- Therefore a direct-contact effect is, or is not, geometrically available.
- Where it is not, these are the mechanisms we cannot assess.

Not supported

- This mutation causes resistance to this drug.
- Binding affinity changes by *X* kcal/mol.
- This drug will still work / will stop working.
- The mutation is *N* Å from *this specific drug* (unless it is the co-crystallised ligand).
- Our pathogenicity score implies anything about drug response.
- Any accuracy figure for our binding-delta prediction.

> THE ONE-SENTENCE POSITION
>
> *"We measure, from experimental structure, whether a mutation can affect a drug by direct contact — and we state plainly when it cannot, and what that leaves unexplained."*

Why this is the stronger pitch

A tool that declines to answer when it cannot is more trustworthy than one that always produces a number. In a clinical-adjacent setting, a confident wrong answer is the failure mode that matters — and the system's central rule already says it: never state a fact you did not retrieve or compute.

We are not reporting a negative result. We are reporting a *correctly scoped* one, with the measurements to back it.

SecondLook Tier 2 · internal technical briefing

Accuracy figures are drawn from the project's own research review and `validation-plan.md`. Distances and deltas are measured outputs from the pipeline's gold-standard run, reproducible via `validation/run_gold_standard.py`.

Pass rate against pre-committed criteria: **0 of 9**. The threshold was not adjusted after seeing results.